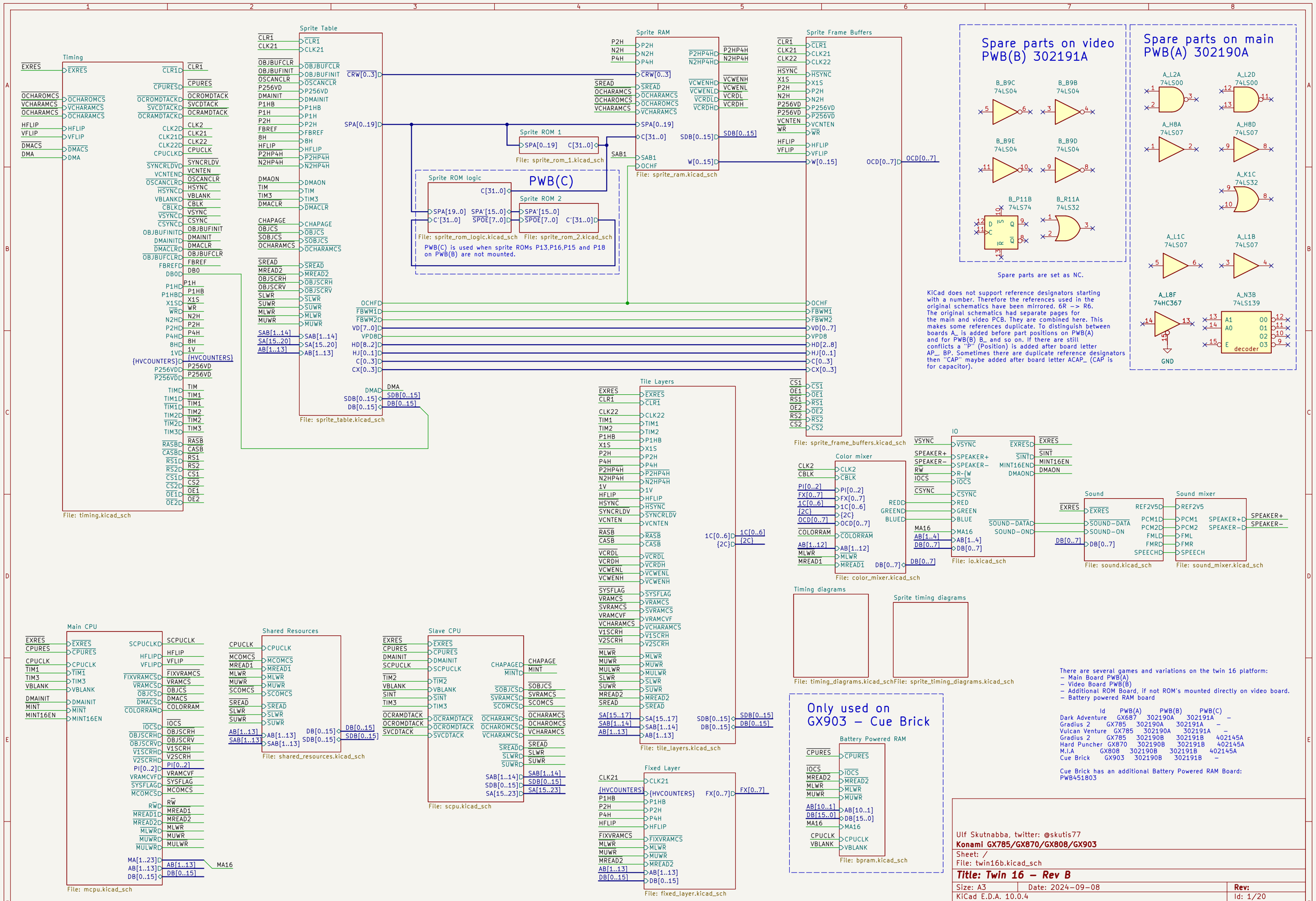
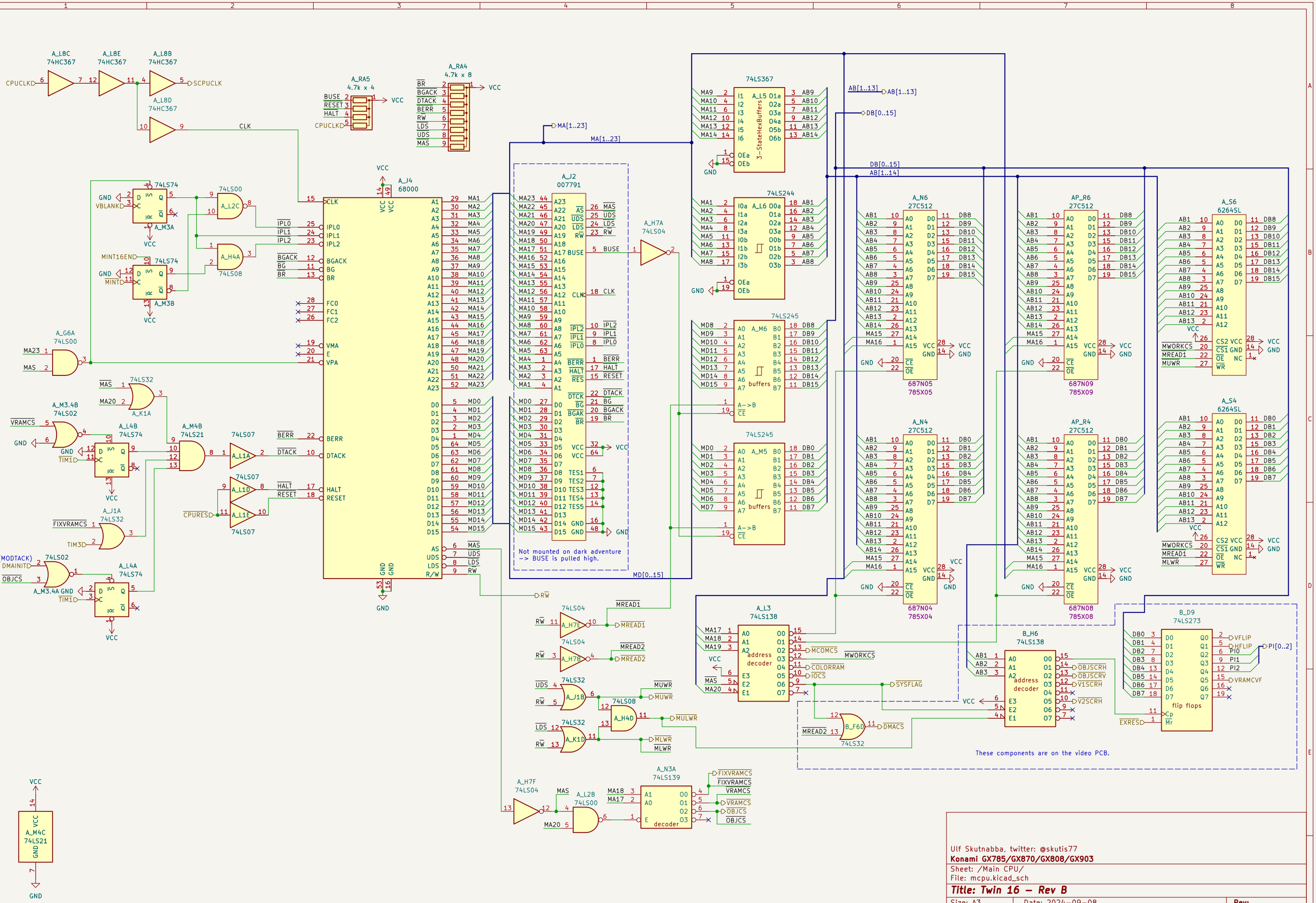

Timing
File: timing.kicad_sch
Sprite Table
File: sprite_table.kicad_sch
Sprite RAM
Sprite ROM 1
File: sprite_rom_1.kicad_sch
Sprite ROM logic
PWB(C)
Sprite ROM 2
File: sprite_rom_logic.kicad_sch
File: sprite_rom_2.kicad_sch
PWB(C) is used when sprite ROMs P13,P16,P15 and P18 on PWB(B) are not mounted.
File: sprite_ram.kicad_sch
Sprite Frame Buffers
File: sprite_frame_buffers.kicad_sch
Spare parts on video PWB(B) 302191A
B_B9C 74LS04
B_B9B 74LS04
B_B9E 74LS04
B_B9D 74LS04
B_P11B 74LS74
B_R11A 74LS32
Spare parts are set as NC.
Spare parts on main PWB(A) 302190A
A_L2A 74LS00
A_L2D 74LS00
A_H8A 74LS07
A_H8D 74LS07
A_K1C 74LS32
A_L1C 74LS07
A_L1B 74LS07
A_L8F 74HC367
A_N3B 74LS139
GND
KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R -> R6. The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).
Tile Layers
File: tile_layers.kicad_sch
Color mixer
File: color_mixer.kicad_sch
IO
File: io.kicad_sch
Sound
File: sound.kicad_sch
Sound mixer
File: sound_mixer.kicad_sch
Timing diagrams
File: timing_diagrams.kicad_sch
Sprite timing diagrams
File: sprite_timing_diagrams.kicad_sch
Main CPU
File: mcpu.kicad_sch
Shared Resources
File: shared_resources.kicad_sch
Slave CPU
File: scpu.kicad_sch
Fixed Layer
File: fixed_layer.kicad_sch
Only used on GX903 – Cue Brick
Battery Powered RAM
File: bpram.kicad_sch
There are several games and variations on the twin 16 platform:
– Main Board PWB(A)
– Video Board PWB(B)
– Additional ROM Board, if not ROM's mounted directly on video board.
– Battery powered RAM board
Id PWB(A) PWB(B) PWB(C)
Dark Adventure GX687 302190A 302191A –
Gradius 2 GX785 302190A 302191A –
Vulcan Venture GX785 302190A 302191A –
Gradius 2 GX785 302190B 302191B 402145A
Hard Puncher GX870 302190B 302191B 402145A
M.I.A GX808 302190B 302191B 402145A
Cue Brick GX903 302190B 302191B –
Cue Brick has an additional Battery Powered RAM Board: PWB451803
Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /
File: twin16b.kicad_sch
Title: Twin 16 – Rev B
Size: A3
Date: 2024-09-08
KiCad E.D.A. 10.0.4
Rev:
Id: 1/20



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Konami GX785/GX870/GX808/GX903

Sheet: /Main CPU/
File: mcpu.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/20

Only used on PWB(A)302190B
Goes to CN2:24.
On PWB(A)302190A CN2:24 is connected to GND.

In the original schematics
DMAINT, MODTACK and DMAINIT are the same

8R and 10R are not mounted when
the GX687 board is used for gradius 2
with GX785 roms.

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Konami GX785/GX870/GX808/GX903

Sheet: /Slave CPU/
File: scpu.kicad_sch

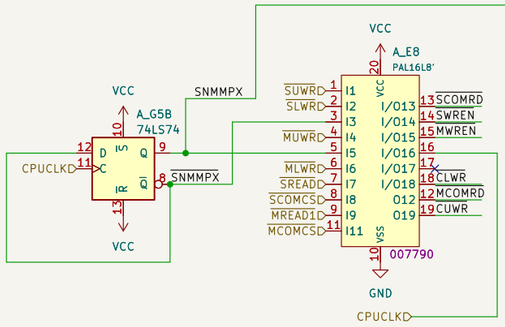
Title: Twin 16 – Rev B

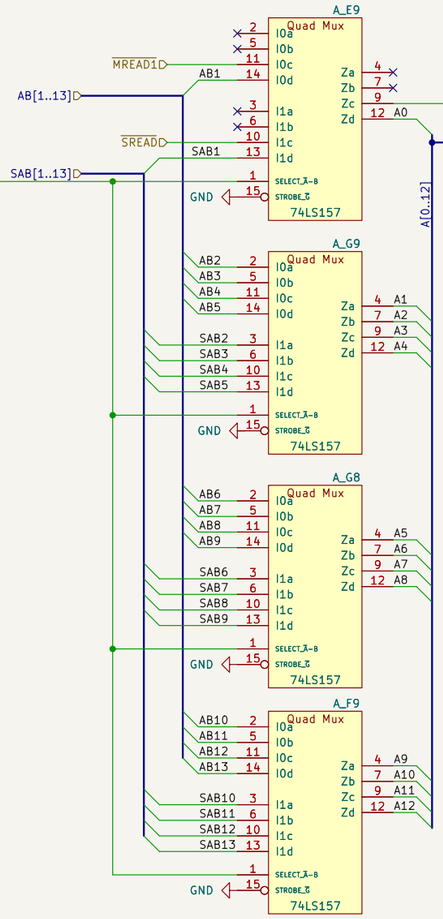
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/20

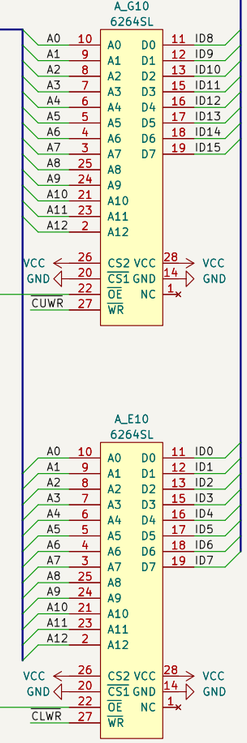
Konami 007790

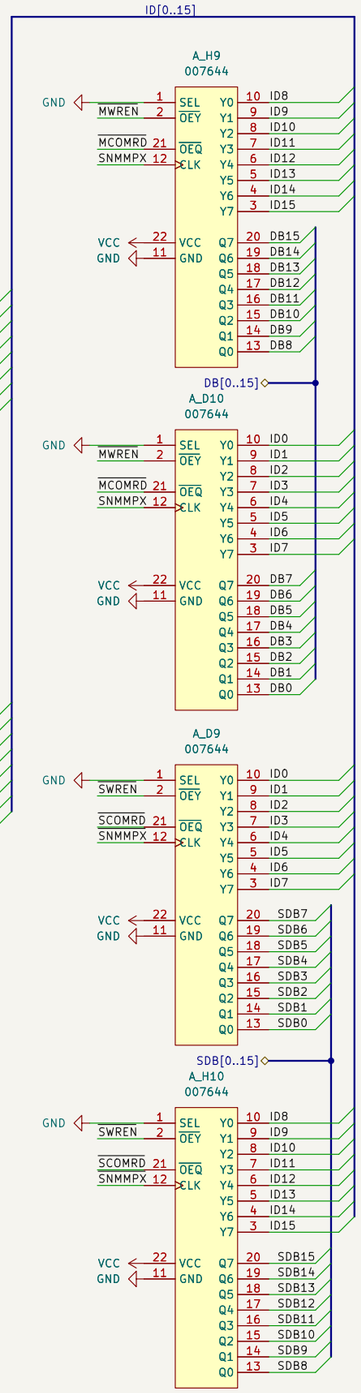
MCOMRD = ~(~MREAD1 & ~MCOMCS) = MREAD1 | MCOMCS
SCOMRD = ~(~SREAD & ~SCOMCS)
SWREN = ~(~SNMMPX & ~SREAD & ~SCOMCS)
MWREN = ~(~SNMMPX & MREAD1 & ~MCOMCS)
CLWR = ~(~SLWR & ~SNMMPX & SNMMPX & ~SREAD & ~SCOMCS & ~CPUCLK | ~SNMMPX & ~MLWR & ~MREAD1 & ~MCOMCS & ~CPUCLK)
CUWR = ~(~SUWR & ~SNMMPX & SNMMPX & ~SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.









This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs
mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

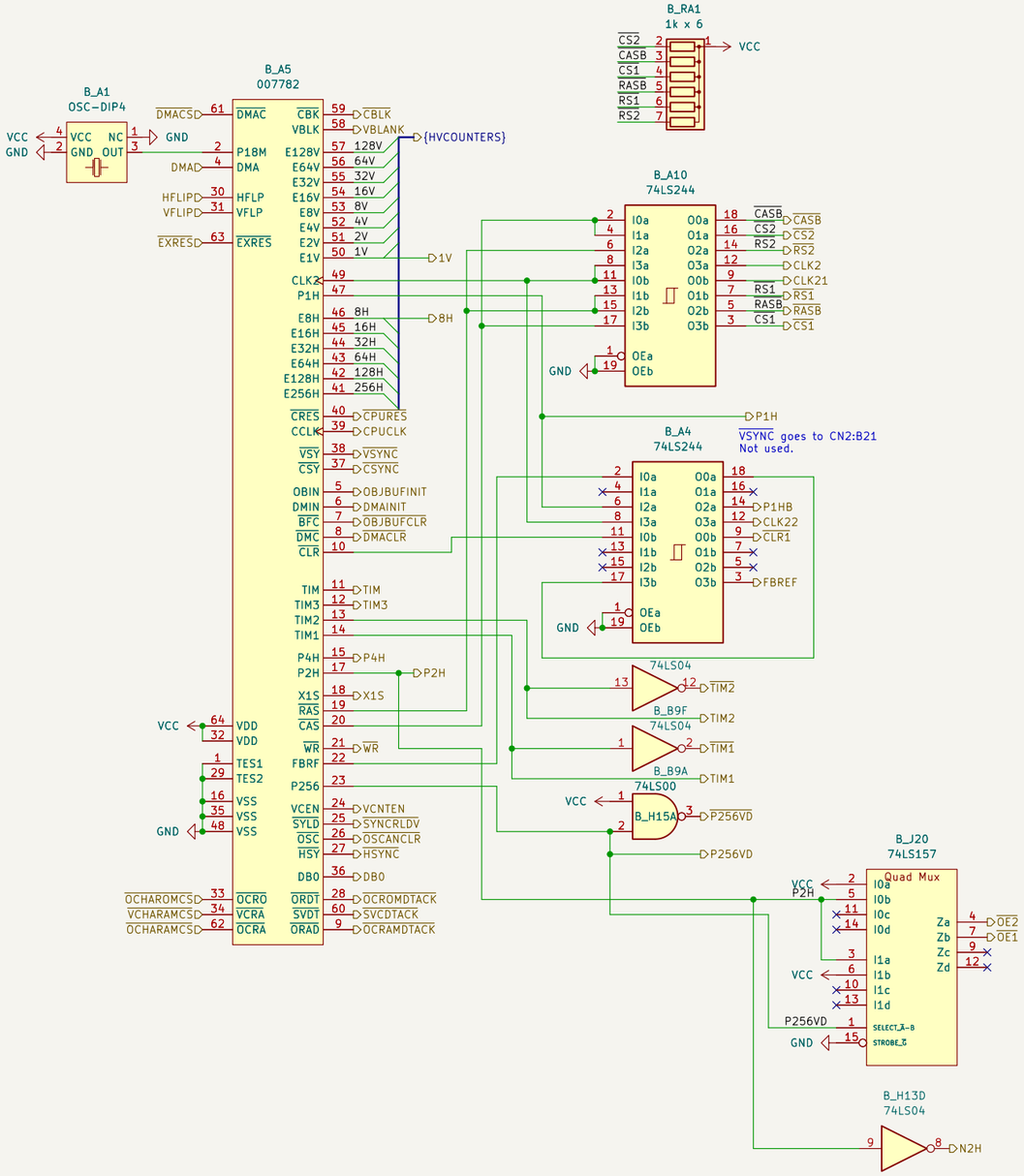
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Konami GX785/GX870/GX808/GX903

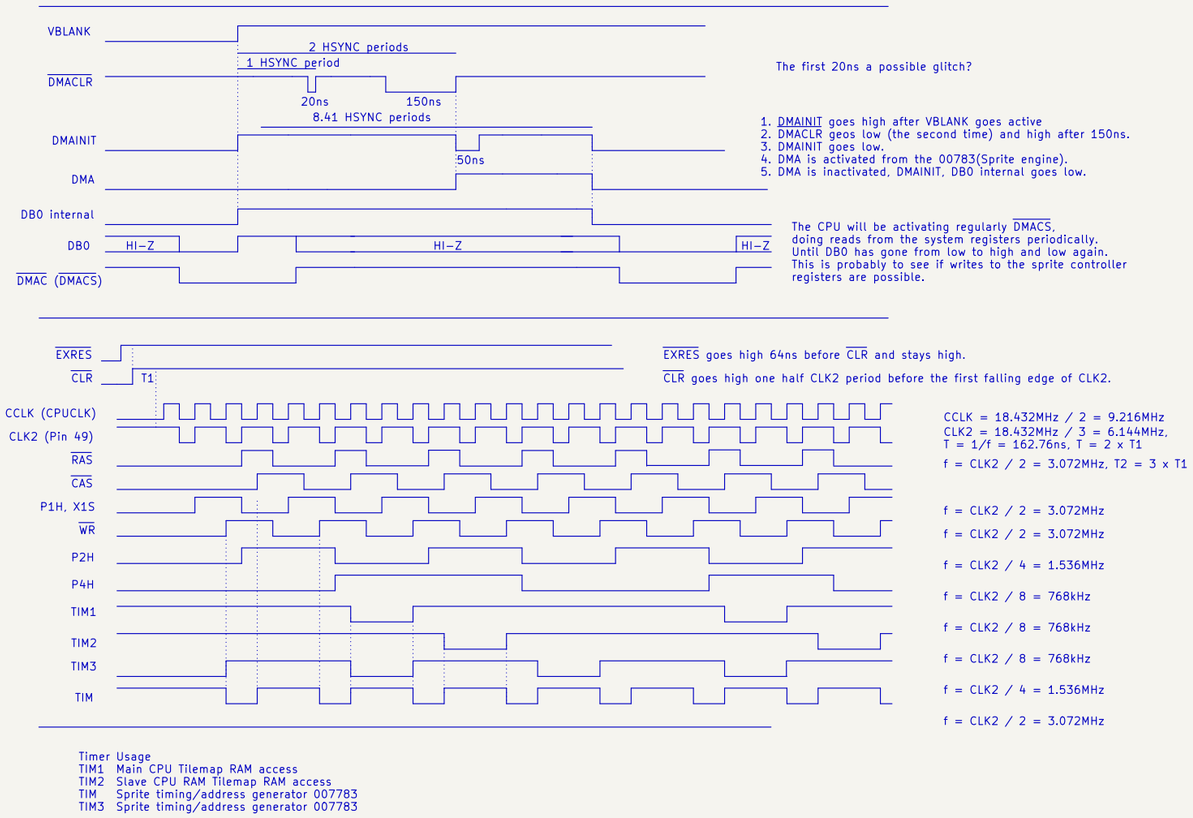
Sheet: /Shared Resources/
File: shared_resources.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/20

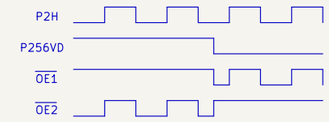


DMA Synchronization

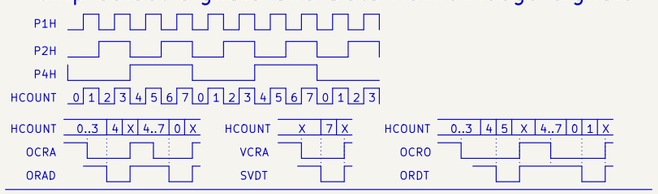


Timer Usage
TIM1 Main CPU Tilemap RAM access
TIM2 Slave CPU RAM Tilemap RAM access
TIM Sprite timing/address generator 007783
TIM3 Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active



Chip select signals and Data Acknowledge signals



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Konami GX785/GX870/GX808/GX903

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/20

B_B3 74LS157 Quad Mux
B_B4 74LS157 Quad Mux
B_B5 74LS157 Quad Mux
B_B6 74LS157 Quad Mux
B_B7 74LS273 flip flops
B_B8 74LS157 Quad Mux
B_C3 6264SL
B_C4 6264SL
B_C5 74LS174 hex flip flop
B_C6 74LS174 hex flip flop
B_C9 74LS273 flip flops
B_C10 74LS273 flip flops
B_D5 74LS174 hex flip flop
B_D6 74LS174 hex flip flop
B_D8 27C128
B_D10 74LS273 flip flops
B_E4A 74LS27
B_E4B 74LS27
B_F6A 74LS32
B_F7 74LS245 buffers
B_F8 74LS245 buffers
B_H13A 74LS04
B_H13F 74LS04
B_A11A 74LS86
B_A11B 74LS86
B_A11C 74LS86
B_A11D 74LS86
B_B10A 74LS86
B_B10B 74LS86
B_B10C 74LS86
B_B10D 74LS86
B_B11A 74LS86
B_B11C 74LS86
B_B11D 74LS86
785G14
870F14
808E14
903E14
7ns delay
20ns delay
10ns delay
50ns delay from OE active to output data valid
Palette, HFLIP and VFLIP is valid for eight pixels at a time and becomes valid on the rising edge of P4H
The total delay for a pixel is seven pixel clocks from when P2H and OE goes high and low respectively.
Two pixels color data is output on each rising edge of P1HB. The current pixel data is multiplexed with P1HB (and HFLIP).
FX[0..3]: Color
FX[4..7]: Palette
ROM
A[0..1] Horizontal position 0..6 in steps of 2
A[2..4] Vertical position 0..7
A[5..13] Tile number
RAM
A[0..5] Horizontal position 0..504 in steps of 8
A[6..10] Vertical position 0..248 in steps of 8
A[11.12] Tilemaps? Up to 4.
D[0..8]: Tile number
D[9..12]: Palette
D13: HFLIP
D14: VFLIP
1. Addresses and OE latched into RAM. This condition sets the start: P2H & P4H = 1.
2. Data out from RAM valid
3. Data out latched into DFF's.
4. DFF's data out valid
4. FX[0..7] Data out valid
5. Palette and color values latched into DFF's 3C and 2E.
6. DFF's data out valid

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Konami GX785/GX870/GX808/GX903

Sheet: /Fixed Layer/
File: fixed_layer.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/20

This is from jammarcade.net

Konami 007644

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

VD[0..12]: Tile number
VD[13..15]: Palette Color

Data latched from the RAM:
Edge Tilemap
T1–T21
T2–T32

It is output 2 pixel clocks later to the ROMs.

When VA is switched to output the bus is updated for each P1H cycle.

The vertical counters are not the same value when they are offset with the scroll registers.

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

Outputs of RAM active when P4H is low, i.e during T1 and T2.

SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

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Konami GX785/GX870/GX808/GX903

Sheet: /Tile Layers/
File: tile_layers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 10.0.4		Id: 7/20

B_B12
007783

B_C13
6264SL

B_C14
6264SL

B_C12
007784

B_F12
007785

74LS245

74LS157
Quad Mux
B_J13

74LS157
Quad Mux
B_J14

74LS157
Quad Mux
B_J18

74LS157
Quad Mux
B_J16

74LS157
Quad Mux
B_J17

B_F6B
74LS32

B_K12A
74LS32

B_A16
PAL16L8'

B_P11A
74LS74

B_RA3
2.2k x 8

B_RA2
2.2k x 8

Each unparsed sprite (object) occupies 40 words in object RAM. Only the first 8 words of each object data is used. The memory range for unparsed data is up 17FFh. That leaves room for 153 (153 x 40 −1 = 17E7h) objects per frame.

On PWB(A) 302190A CHAPAGE (CN2:A23) is connected to GND

007784 Sprite parser

− subtracts the scroll value for the sprites from current coordinates for the sprites.
− When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x− and y−scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
− Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21′ = CLK21

$\overline{LRW}$ = -($\overline{DMAONS}$ & OBJBUFINIT & CLK21′ |
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP | 8H & ~HFLIP)
~LWR & ~DMAINIT)

$\overline{URW}$ = -($\overline{DMAONS}$ & OBJBUFINIT & CLK21′ |
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP | ~8H & HFLIP) |
~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

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Konami GX785/GX870/GX808/GX903

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16 − Rev B

Size: A3	Date: 2024−09−08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/20

Graphics ROM and RAM

ROM naming

	ID	P13	P15	P16	P18
cuebrickj	GX903	NP	NP	NP	NP
miaj	GX808	808D15	NP	808D17	NP
hpuncher	GX870	870C15	870C16	870C17	870C18
gradius2 / vulcan	GX785	785F15	785F16	785F17	785F18

NP = Not Populated
On some boards P13,P15,P16 and P18 are NP and replaced with PWB(C) 402145A. See separate schematics.

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
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Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Konami 007788

The inverted ~P2H is written as N2H.

~{OCRDL} = ~(~{SREAD} & SAB1 & ~{OCHARAMCS} & {OCHAROMCS} | ~{SREAD} & SAB1 & ~{OCHAROMCS})
~{OCWL} = ~(~P2H & SREAD & SAB1 & ~{OCHARAMCS})
~{OCRDH} = ~(~{SREAD} & ~SAB1 & ~{OCHARAMCS} & {OCHAROMCS} | ~{SREAD} & ~SAB1 & ~{OCHAROMCS})
~{OCWH} = ~(~P2H & SREAD & ~SAB1 & ~{OCHARAMCS})
~{VCWENL} = ~(P2H & P4H & ~{VCHARAMCS} & SREAD & SAB1)
~{VCRDL} = ~(~{VCHARAMCS} & ~{SREAD} & SAB1)
~{VCRDH} = ~(~{VCHARAMCS} & ~{SREAD} & ~SAB1)
~{VCWENH} = ~(P2H & P4H & ~{VCHARAMCS} & SREAD & ~SAB1)
~{P2HP4H} = ~(P2H & P4H)
N2HP4H = ~(~P2H & P4H)

Sprite Character Banks

Page	SPA19	SPA18	SPA17
1	X	L	L
2	X	L	H
3	L	H	L
4	H	H	L
5 RAM	X	H	H

Pixel color data output rate is:
16–bits / 2 pixels = 8 bits / pixel.

W[15]: Vertical flip

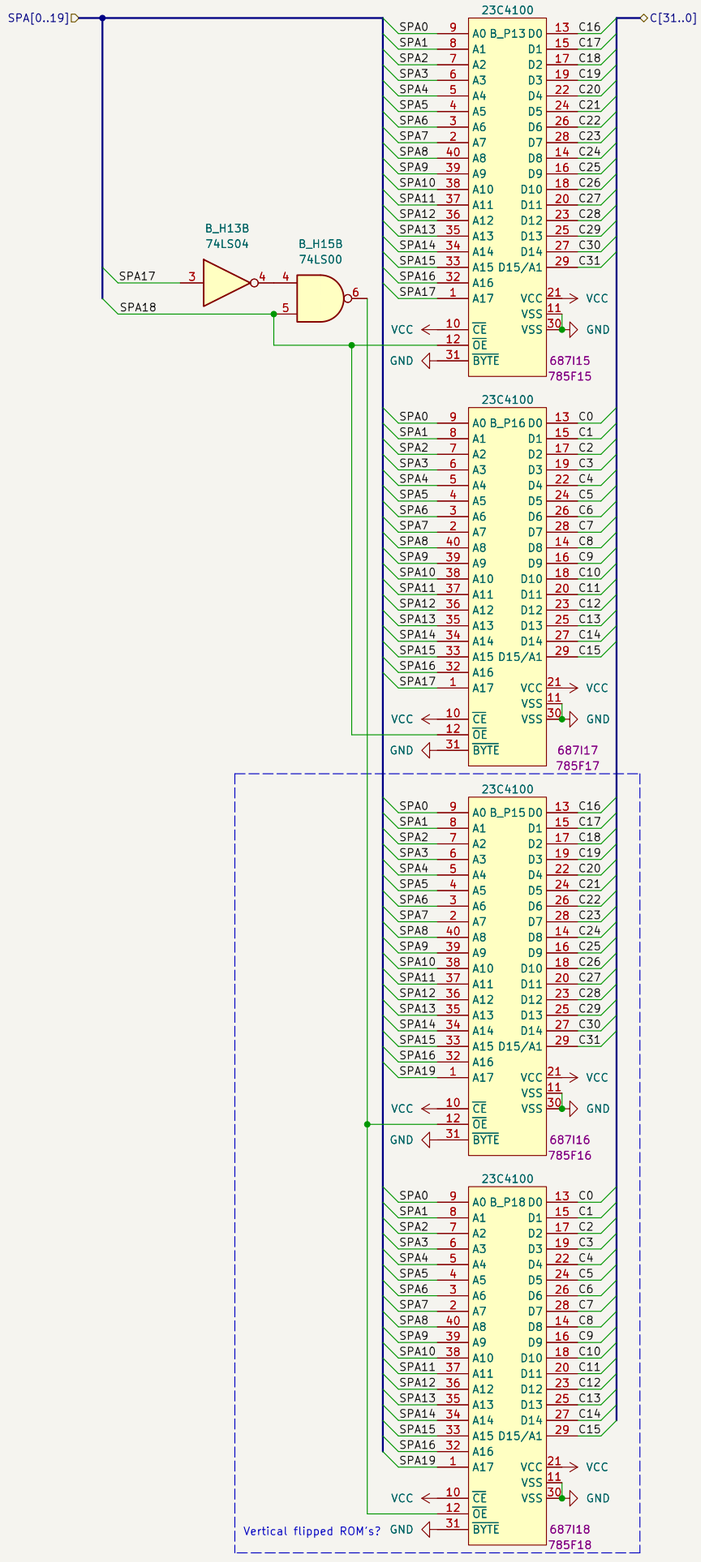
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite RAM/
File: sprite_ram.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024–09–08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/20



Sheet: /Sprite ROM 1/	
File: sprite_rom_1.kicad_sch	
Title: Twin 16 – Rev B	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 10.0.4	Id: 10/20
	Rev:

C_A14
4.7k x 8
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7
VCC

C'[31..0]

C_A15
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C0 C1 C2 C3 C4 C5 C6 C7
OEa OEb
74LS244
VCC GND

C[31..0]

SPA[19..0]

C_C12A
74LS08
SPA17 SPA18
3

CR_D8
4.7k x 8
SPA7 SPA6 SPA5 SPA4 SPA3 SPA2 SPA1 SPA0
VCC

C_D9
SPA0 SPA1 SPA2 SPA3 SPA4 SPA5 SPA6 SPA7
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
SPA'0 SPA'1 SPA'2 SPA'3 SPA'4 SPA'5 SPA'6 SPA'7
GND
OEa OEb
74LS244

C_D10
4.7k x 8
C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15
VCC

C_D11
C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C8 C9 C10 C11 C12 C13 C14 C15
OEa OEb
74LS244
VCC GND

C_C8
4.7k x 8
SPA9 SPA10 SPA8 SPA11 SPA12 SPA15 SPA13 SPA14
VCC

C_D9
SPA8 SPA9 SPA10 SPA11 SPA12 SPA13 SPA14 SPA15
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
SPA'8 SPA'9 SPA'10 SPA'11 SPA'12 SPA'13 SPA'14 SPA'15
GND
OEa OEb
74LS244

SPA'[15..0]

C_C10
4.7k x 8
C'16 C'17 C'18 C'19 C'20 C'21 C'22 C'23
VCC

C_C11
C'19 C'17 C'18 C'16 C'20 C'21 C'22 C'23
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C19 C17 C18 C16 C20 C21 C22 C23
OEa OEb
74LS244
VCC GND

Mirrored when connected to PWB(B)

C_CN3
SPA9 SPA8 SPA7 SPA6 SPA5 SPA4 SPA3 SPA2 SPA1 SPA0
GND
VCC
C15 C14 C13 C12 C11 C10 C9 C8 C7 C6 C5 C4 C3 C2 C1 C0
SPA10 SPA11 SPA12 SPA13 SPA14 SPA15 SPA16 SPA17 SPA18 SPA19
C31 C30 C29 C28 C27 C26 C25 C24 C23 C22 C21 C20 C19 C18 C17 C16
DIN64PS

C_B14
4.7k x 8
C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31
VCC

C_B15
C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31
I0a I1a I2a I3a I0b I1b I2b I3b
O0a O1a O2a O3a O0b O1b O2b O3b
C31 C30 C29 C28 C27 C26 C25 C24
OEa OEb
74LS244
VCC GND

SPOE[7..0]

C_D12
4.7k x 4
SPA16 SPA17 SPA18
VCC

C_D13
74LS138
SPA16 SPA17 SPA18
A0 A1 A2
address decoder
E3 E2 E1
VCC GND
O0 O1 O2 O3 O4 O5 O6 O7
SPOE0 SPOE1 SPOE2 SPOE3

SPA19

C_C15B
74LS32
SPOE4

C_C13A
74LS04
SPA19

C_C15A
74LS32
SPOE5

C_C15C
74LS32
SPOE7

C_C15D
74LS32
SPOE6

C_C12B
74LS08

C_C12C
74LS08

C_C12D
74LS08

C_C13B
74LS04

C_C13C
74LS04

C_C13D
74LS04

C_C13E
74LS04

C_C13F
74LS04

PWB(C) 402145A
Konami GX785/GX870/GX808/GX903
Sheet: /Sprite ROM logic/
File: sprite_rom_logic.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 11/20

C_B12 — 27C512 — F17B

C_B11 — 27C512 — F17D

C_B10 — 27C512 — F17F

C_B9 — 27C512 — F17H

C_A12 — 27C512 — F18B

C_A11 — 27C512 — F18D

C_A10 — 27C512 — F18F

C_A9 — 27C512 — F18H

C_D8 — 27C512 — F17A

On the PCB the position is D9

C_D7 — 27C512 — F17C

C_D6 — 27C512 — F17E

C_D5 — 27C512 — F17G

C_D4 — 27C512 — F18A

C_D3 — 27C512 — F18C

C_D2 — 27C512 — F18E

C_D1 — 27C512 — F18G

C_B8 — 27C512 — F15B

C_B7 — 27C512 — F15D

C_B6 — 27C512 — F15F

C_B5 — 27C512 — F15H

C_B4 — 27C512 — F16B

C_B3 — 27C512 — F16D

C_B2 — 27C512 — F16F

C_B1 — 27C512 — F16H

C_A8 — 27C512 — F15A

C_A7 — 27C512 — F15C

C_A6 — 27C512 — F15E

C_A5 — 27C512 — F15G

C_A4 — 27C512 — F16A

C_A3 — 27C512 — F16C

C_A2 — 27C512 — F16E

C_A1 — 27C512 — F16G

$\overline{SPOE}$[7..0]

SPA'[15..0]

C'[31..0]

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Sheet: /Sprite ROM 2/

File: sprite_rom_2.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 11/20

Frame buffer 1

Frame buffer 2

Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.

Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

OCD[0:3]: Pixel color
OCD[4:7]: Palette

Frame buffer 1 Palette color

Frame buffer 2 Palette color

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Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 10.0.4		Id: 13/20

Components are on Video PCB

Components are on Main PCB

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = 1C |
PI0 & ~PI1 & FX |
~PI1 & OCD0 & OCD1 & OCD2 & OCD3 & FX |
~PI1 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & FX |
PI1 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3 |
PI1 & ~2C0 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & ~2C1 & ~2C2 & ~2C3

~SELB = ~1C & OCD0 & OCD1 & OCD2 & OCD3 |
~1C & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 |
~1C & PI0 & ~PI1 & FX |
~1C & PI0 & PI1 & 2C0 & BUS |
~1C & PI0 & PI1 & BUS & 2C1 |
~1C & PI0 & PI1 & BUS & 2C2 |
~1C & PI0 & PI1 & BUS & 2C3

~SHADOW = ~1C & ~PI1 & OCD0 & OCD1 & OCD2 & OCD3 |
~1C & ~PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~FX |
~1C & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~BUS |
~1C & PI0 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3

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Konami GX785/GX870/GX808/GX903		
Sheet: /Color mixer/		
File: color_mixer.kicad_sch		
Title: Twin 16 – Rev B		
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 14/20

A_SW1
SW_DIPS_08
A_RA1
2.2k x 8
SW[11..18]
A_T4
74LS257
A_SW2
SW_DIPS_08
A_RA2
2.2k x 8
SW[21..28]
A_03
74LS257
A_SW3
SW_DIPS_04
A_RA3
2.2k x 4
SW[31..34]
A_U4A
74LS367
A_T5
74LS138
address decoder
AB[1..4]
R-{WD}
VCC
GND
SOUND-DATA
AFR
SHOOT
IOCS
MA16
PWR_FLAG
A_L1F
74LS07
A_R2
470
A_R1
100
CSYNC
JAMMA_CONN
COIN_CNT_1
COIN_CNT_2
LOCK_COIL_1
LOCK_COIL_2
SPEAKER_(+)
SPEAKER_(-)
NOT_USED
RED
GREEN
BLUE
SYNC
VIDEO_GND
SERVICE
TEST
TILT
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
1P_BTN_4
2P_BTN_4
1P_SELECT
2P_SELECT
COINCOUNTER1
COINCOUNTER2
SERVICE1
COIN1
COIN2
1PSTART
2PSTART
1PUP
2PUP
1PDOWN
2PDOWN
1PLEFT
2PLEFT
1PRIGHT
2PRIGHT
1PSHOOT1
2PSHOOT1
1PSHOOT2
2PSHOOT2
1PSHOOT3
2PSHOOT3
A_M2
74LS253
A_N2
74LS253
A_O2
74LS253
A_PP2
74LS253
Y[3..32]
A_CRA1
005273
A_CRA2
005273
A_CRA3
005273
A_CRA4
005273
A_CRA5
005273
A_CRA6
005273
A_CRA7
005273
A_CRA8
005273
A_U5
74LS273
flip flops
DB[0..7]
SOUND-ON
SINT
MINT16EN
DMAON
GAMEEND
A_E1
005924
JP
AFR
WDRES
EXRES
A_RA6D
4.7k x 4
A_L8A
74HC367
R8
4.7k
D1
10D1
TR2
2SC2320
TR1
2SD612
CN3
15P NH CONNECTOR
CN9
VSYNC
GAMEEND
4P SV CONNECTOR

AFR resets the wathdog timer

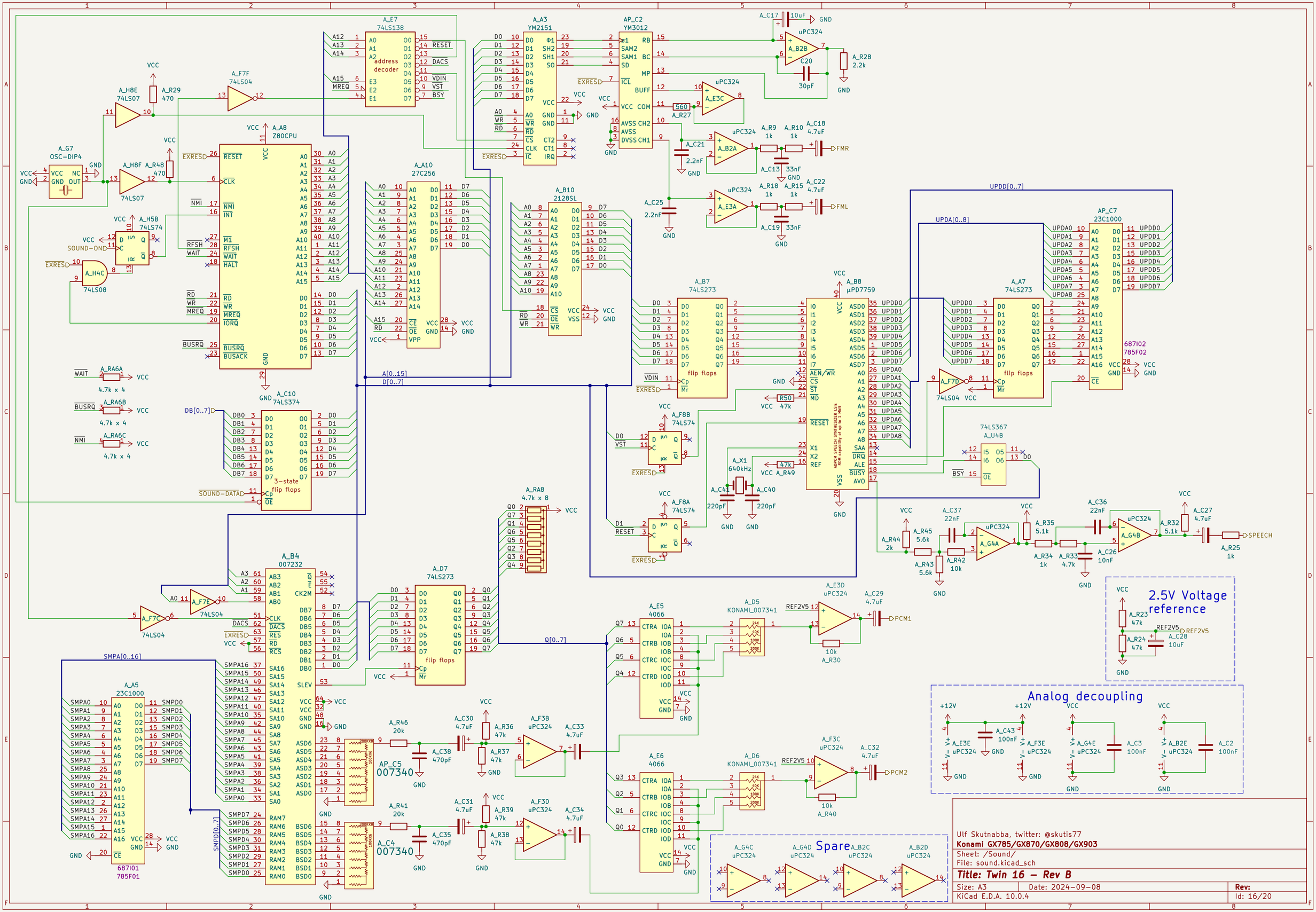
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Konami GX785/GX870/GX808/GX903

Sheet: /IO/
File: io.kicad_sch

Title: Twin 16 – Rev B

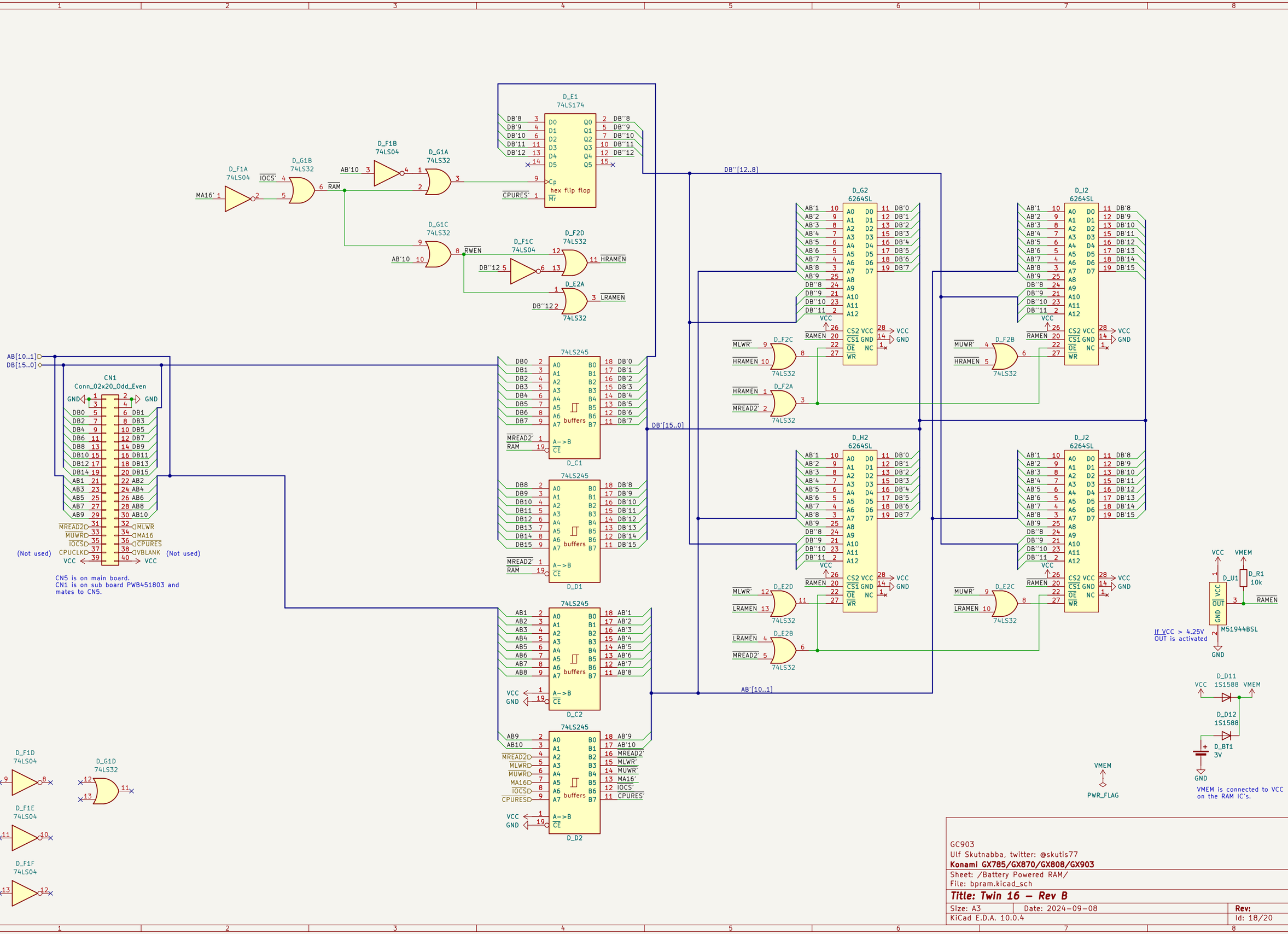
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 15/20


A_E7
74LS138
address decoder
A_A3
YM2151
AP_C2
YM3012
uPC324
A_B2B
A_E3C
A_B2A
A_E3A
A_A8
Z80CPU
A_A10
27C256
A_B10
2128SL
A_B7
74LS273
flip flops
A_B8
µPD7759
ADPCM SPEECH SYNTHESIZER LSIs
ROM capability of up to 1 Mbit
A_A7
74LS273
AP_C7
23C1000
687102
785F02
A_F7F
74LS04
A_H8E
74LS07
A_H8F
74LS07
A_G7
OSC-DIP4
A_H5B
74LS74
A_H4C
74LS08
A_F7E
74LS04
A_F7C
74LS04
A_F7D
74LS04
A_F8B
74LS74
A_F8A
74LS74
74LS367
A_U4B
A_C10
74LS374
3-state
flip ops
A_B4
007232
AP_C5
007340
A_C4
007340
A_A5
23C1000
687101
785F01
A_RA8
4.7k x 8
A_RA6A
A_RA6B
A_RA6C
4.7k x 4
A_X1
640kHz
A_D7
74LS273
A_D5
KONAMI_007341
A_D6
KONAMI_007341
A_E5
4066
A_E6
4066
A_E3D
uPC324
A_F3C
uPC324
A_F3B
uPC324
A_F3D
uPC324
A_G4A
uPC324
A_G4B
uPC324
A_G4C
uPC324
A_G4D
uPC324
A_B2C
uPC324
A_B2D
uPC324
Spare
2.5V Voltage reference
Analog decoupling
A[0..15]
D[0..7]
SMPA[0..16]
SMPD[0..7]
UPDA[0..8]
UPDD[0..7]
Q[0..7]
DB[0..7]
SPEECH
PCM1
PCM2
FMR
FML
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Konami GX785/GX870/GX808/GX903
Sheet: /Sound/
File: sound.kicad_sch
Title: Twin 16 – Rev B
Size: A3
Date: 2024-09-08
KiCad E.D.A. 10.0.4
Rev:
Id: 16/20

To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

A_R17
1k
FMRD
A_R12
7.5k
SPEECHD
A_R11
5.1k
PCM1D
A_R19
5.1k
PCM2D
A_R16
1k
FMLD
A_R22
7.5k
SPEECHD
A_R31
5.1k
PCM1D
A_R20
5.1k
PCM2D
A_E3B
uPC324
REF2V5D
A_R13
12k
+12V
A_R52
3.3k
A_C15
4.7uF
A_R14
10k
A_F3A
uPC324
REF2V5D
A_R21
12k
+12V
A_R53
3.3k
A_C16
4.7uF
A_R26
10k
CN6
4P NH CONN
GND
CN7
4P NH CONN
APOT_R53
1k
APOT_R52
1k
A_C11
100nF
A_C12
100nF
GND
A_C9
100uF
ACAP_C10
100uF
A_D1
MB3722
IN-A
IN+A
IN+A
IN-B
VCC
GNDIN
GNDOUT
BYPASS
BSA
OUTA
BSB
OUTB
+12V
ACAP_C3
1000uF
A_C8
100uF
A_C7
100uF
A_C14
10uF
ACAP_C2
1000uF
A_C6
100nF
A_R6
1
A_C5
100nF
A_R7
1
A_C1
1000uF
SPEAKER-
SPEAKER+
CN8
4P NH CONN

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Konami GX785/GX870/GX808/GX903
Sheet: /Sound mixer/
File: sound_mixer.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 17/20



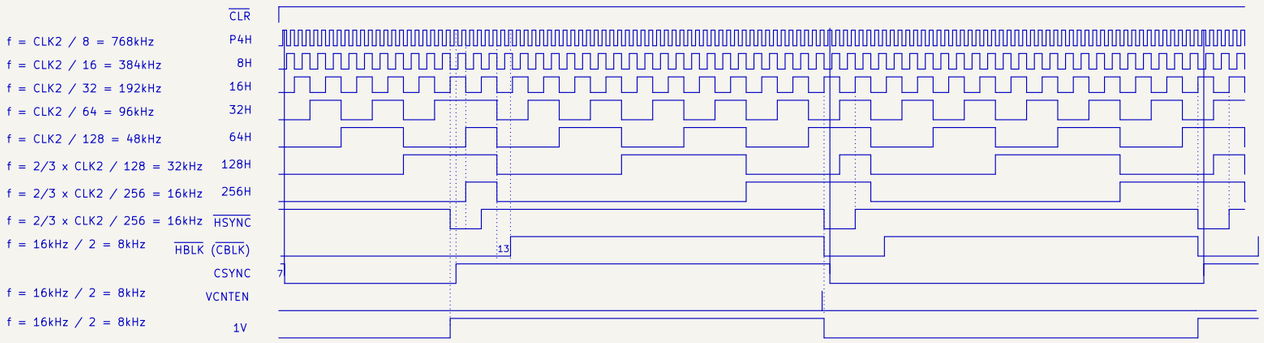
GC903
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Konami GX785/GX870/GX808/GX903

Sheet: /Battery Powered RAM/
File: bpram.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 18/20

Horizontal signals

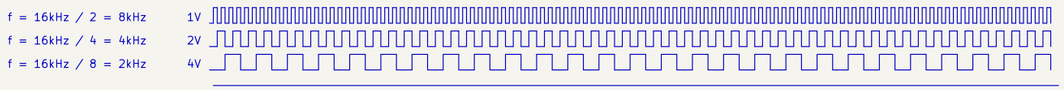


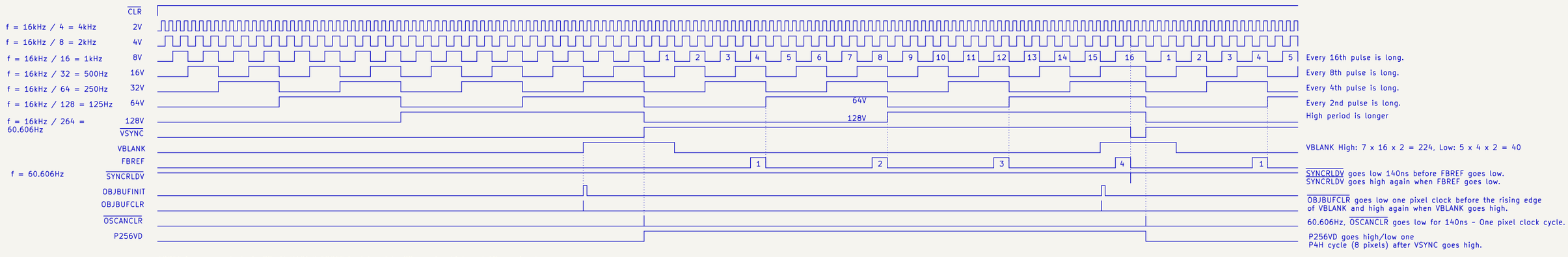
The first HSYNC is at HCOUNT 175 after reset.

– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

Horizontal and vertical synch timing diagrams

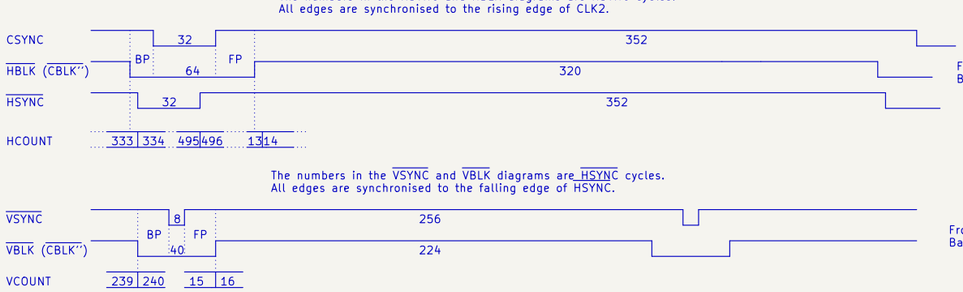
Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.

Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

CBLK'' is at the output of color mixer 007593.
CBLK' is shifted inside the 007593 one pixel clock.

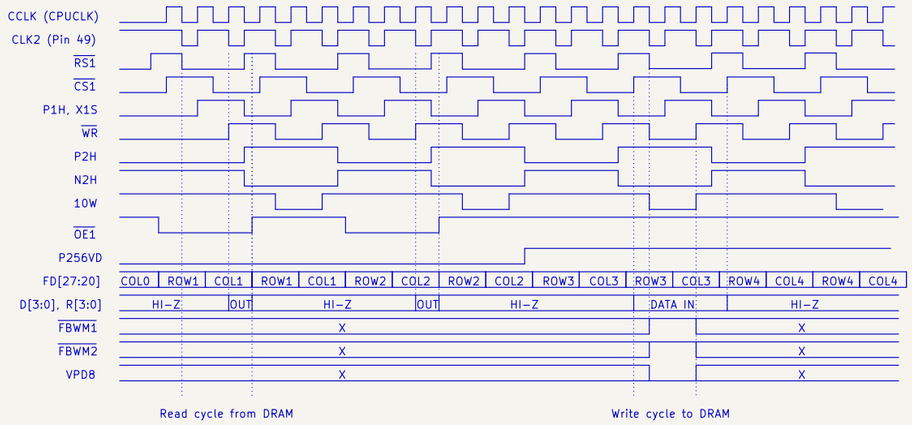
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Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 19/20



The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 10.0.4		Id: 20/20